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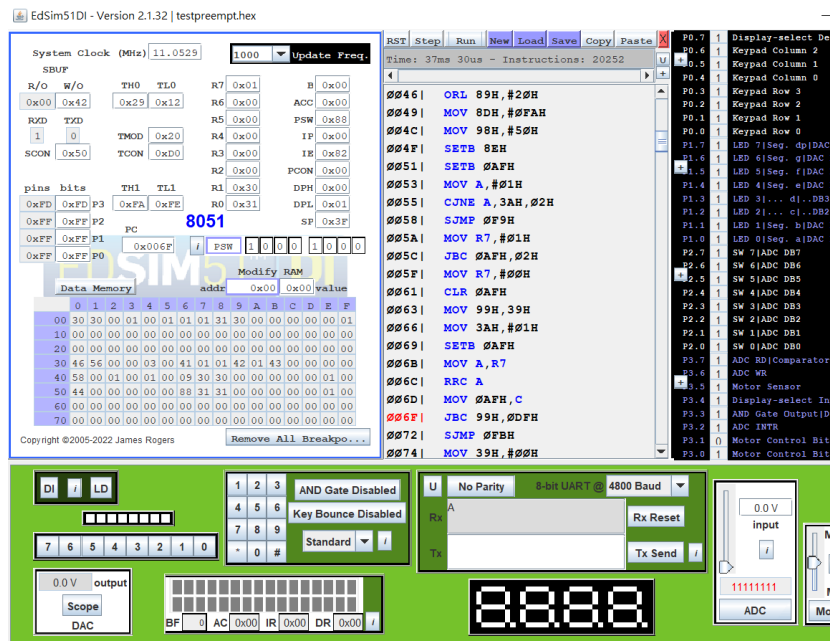
Before calling ThreadCreate(main):

The screenshot shows the EdSim51DI Version 2.1.32 interface. The System Clock is set to 11.0529 MHz. The PC register is 8051. The Data Memory window shows the initial state of memory. The instruction list on the right shows the initial instructions: RET, LJMPL 01F7H, RETI, MOV 89H, #00H, MOV 0A8H, #82H, SETB 8CH, MOV 34H, #00H, MOV 37H, #00H, MOV A, #0FCH, ADD A, 37H, JC 0EH, MOV A, 37H, ADD A, #30H, MOV R0, A, MOV @R0, #00H, MOV A, 37H, INC A, MOV 37H, A, SJMP 0ECH, MOV DPTR, #0074H, and LCALL 00CBH. The hardware components at the bottom include a keypad, a display, a scope, a DAC, and an ADC.

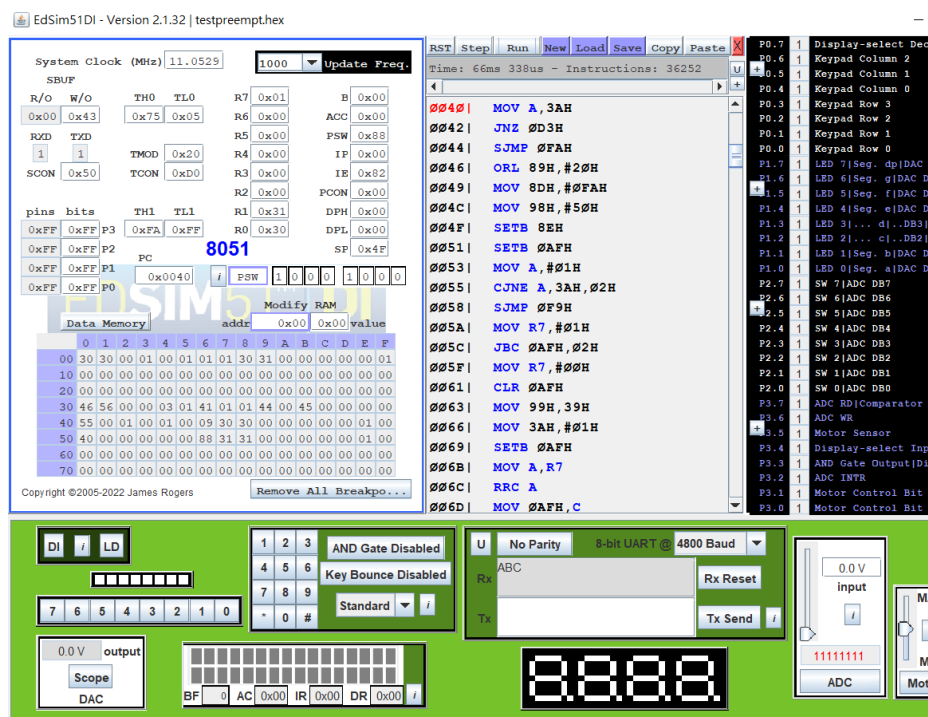
Before calling ThreadCreate(Produder):

The screenshot shows the EdSim51DI Version 2.1.32 interface. The System Clock is set to 11.0529 MHz. The PC register is 8051. The instruction list on the right shows the initial instructions: LCALL 00CBH, LJMPL 0046H, LJMPL 008EH, RET, RET, RET, RET, LJMPL 01F7H, RETI, MOV 89H, #00H, MOV 0A8H, #82H, SETB 8CH, MOV 34H, #00H, MOV 37H, #00H, MOV A, #0FCH, ADD A, 37H, JC 0EH, MOV A, 37H, ADD A, #30H, MOV R0, A, MOV @R0, #00H, and MOV 0R0, #00H. The hardware components at the bottom include a keypad, a display, a scope, a DAC, and an ADC.

After calling ThreadCreate(), 7 objects has been pushed into the stack of the thread.



Value in address 0x35 is 0, means that thread 0 is currently running. So Consumer is running.



Value in address 0x35 is 1, means that thread 1 is currently running. So Producer is running.

Since the thread switching process is triggered by the interrupt, if it is not triggering on a regular basis, then the characters couldn't be outputted successfully.